IEEE PHM 2014 Fuel Cell Prognostics Data Challenge

Technical Dataset and Experimental Description

1. Overview

The IEEE PHM 2014 Data Challenge was organized by the IEEE Reliability Society in collaboration with the FCLAB Research Federation, FEMTO-ST Institute, and the Labex ACTION program. The primary objective of the challenge was to advance prognostics and health management (PHM) methodologies for Proton Exchange Membrane Fuel Cell (PEMFC) systems, with particular emphasis on remaining useful life (RUL) prediction.

Participants from both academic and industrial sectors were invited to develop robust data-driven or physics-informed approaches capable of estimating the health state and predicting the future degradation behavior of PEM fuel cell stacks. High-performing teams were recognized and invited to present their work at the IEEE International Conference on Prognostics and Health Management.

2. Motivation and Scope

Although PEM fuel cell systems are considered promising candidates for clean energy conversion, their commercial deployment remains limited by durability concerns. Typical PEMFC lifetimes are on the order of 2,000 operating hours, which is insufficient for many real-world applications such as transportation systems that require lifetimes exceeding 6,000 hours.

Improving durability requires not only understanding degradation mechanisms but also developing reliable prognostic models capable of forecasting performance degradation and estimating RUL under realistic operating conditions. The IEEE PHM 2014 Data Challenge was designed to support this objective by providing high-quality experimental datasets collected under controlled ageing conditions.

3. Experimental Test Bench

All experiments were conducted at FCLAB using a dedicated fuel cell test platform designed for stacks with electrical power ratings up to 1 kW. The test bench allowed precise monitoring and control of key operating parameters, including:

- Stack temperature and cooling water flow

- Hydrogen and air flow rates and pressures
- Gas humidification levels
- Stack and individual cell voltages
- Operating current and current density

Gas humidification was achieved using independent boiler systems for hydrogen and air streams, while thermal regulation of the stack was maintained via a water-cooling loop. Electrical loading was controlled using an active electronic load to ensure accurate current regulation.

4. Fuel Cell Stack Configuration

The fuel cell stacks used in the challenge were assembled at FCLAB and consisted of five cells connected in series. Each cell had an active area of 100 cm^2 and employed commercially available membranes, gas diffusion layers, and machined bipolar plates. The nominal operating current density was 0.7 A/cm^2, with a maximum current density of 1.0 A/cm^2.

5. Ageing Test Protocols

Two independent long-term durability tests were performed to generate the challenge datasets.

5.1 FC1: Stationary Operating Conditions

The first stack (FC1) was operated under constant current conditions at 70 A, corresponding to a current density of 0.7 A/cm^2. This test served as a reference ageing experiment. Periodic characterization was performed approximately every 160 hours and included:

- Electrochemical Impedance Spectroscopy (EIS)
- Polarization curve measurements
- Long-term voltage monitoring

Characterizations were conducted at predefined ageing times up to approximately 1,000 hours.

5.2 FC2: Dynamic Operating Conditions

The second stack (FC2) was subjected to dynamic electrical loading to simulate realistic operating conditions. A high-frequency triangular ripple current (±10% of nominal current at 5 kHz) was superimposed on the base current of 70 A. Similar characterization procedures to FC1 were conducted at regular intervals.

6. Dataset Description

The datasets provided for the challenge include both time-domain monitoring data and frequency-domain characterization data:

6.1 Monitoring Data

Continuous monitoring data include measurements such as:

- Individual cell and stack voltages
- Operating current and current density
- Gas inlet and outlet temperatures
- Gas pressures and flow rates
- Cooling water parameters
- Air inlet humidity

These data capture the temporal evolution of the fuel cell stack during long-term operation and directly reflect degradation behavior.

6.2 Characterization Data

Characterization datasets consist of:

- Polarization curves, representing static voltage–current behavior
- Electrochemical impedance spectroscopy (EIS) measurements, providing insight into internal electrochemical processes and resistance components

7. Challenge Structure

The IEEE PHM 2014 Data Challenge was divided into two independent tasks:

7.1 State of Health Estimation

Participants were required to estimate the real and imaginary components of stack impedance at selected frequencies and ageing times for FC2. Performance was evaluated using Euclidean distance between predicted and actual impedance values.

7.2 Remaining Useful Life (RUL) Prediction

RUL was defined as the remaining operating time before the fuel cell stack experienced a predefined percentage loss in output power. Predictions were made at a reference time of 550 hours for FC2. Scoring penalized late predictions more severely than early ones to encourage conservative prognostic estimates.

8. Data Organization and Format

All datasets were distributed in compressed archives containing ASCII-formatted CSV files. Separate files were provided for monitoring data, polarization curves, and EIS measurements. Each file followed a strict column structure to ensure consistency across datasets.

9. Summary

The IEEE PHM 2014 Data Challenge provides a comprehensive and well-structured benchmark dataset for evaluating prognostic algorithms applied to PEM fuel cell systems. By combining long-term monitoring data with detailed electrochemical characterization under both static and dynamic operating conditions, the dataset enables rigorous development and validation of degradation modeling and RUL prediction techniques.